# 02. Problem Statement: Core Challenges in AI Agent Systems

1. **Monolithic LLM Bottleneck & High Cost:** Traditional platforms route all planning, reasoning, and tool use through a single massive LLM, causing severe prompt bloat, high latency, and $3\times$ higher inference expenses.
   ➔ **PrismSpace Solution:** Dynamically routes sub-tasks to cost- and latency-optimal models (Groq for rapid inference, OpenAI/NIM for complex reasoning).

2. **Sequential & Rigid Task Execution:** Existing agent chains run strictly in linear sequence; independent sub-tasks wait needlessly in line, and a single step crash causes total pipeline collapse.
   ➔ **PrismSpace Solution:** Deconstructs goals into dynamic Directed Acyclic Graphs (DAGs) to execute independent tasks in parallel ($K \leq 3$) with isolated fault recovery.

3. **Fragmented Tool Silos & Black-Box Runs:** Proprietary integrations isolate essential productivity tools (Gmail, Drive, Terminal, Local Files), while users remain blind during long runs with no status logs.
   ➔ **PrismSpace Solution:** Standardizes all tool connections via Model Context Protocol (MCP) paired with real-time execution telemetry and progress logs.

4. **Safety Hazards & Cloud Data Lock-in:** Autonomous agents execute irreversible actions (sending emails, modifying files) without oversight, while cloud databases expose private user data.
   ➔ **PrismSpace Solution:** Enforces Human-in-the-Loop (HITL) approval gates, ORPO alignment, and zero-latency local browser database storage (`PrismDB`).